How do you like your dairy?

A review of how the world tastes milk products

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ABSTRACT

How you perceive and consume dairy is determined by various factors. One of these factors is culture. This paper looks at the relationship between dairy products and their country of origin based on textual descriptions. To do this, Wikipedia pages are used to find dairy products and their origin. Moreover, descriptors are defined to find possible relationships between product and country. The results are validated using human judgement and other sources, and show that the Wikipedia-derived dataset is skewed towards Europe and Asia (with India and Turkey contributing the most products). Moreover, *fermented* and *dessert* are the most common processing and serving descriptors. Lastly, some descriptors are geographically widespread (e.g., *fermented*, *dessert*) whereas others are more localized in the descriptor maps (e.g., *cooked* and *buffalo*). Manual checks indicate descriptor extraction is noisier for serving/processing than for animal source.

Code available at: https://github.com/dewi-elisa/dairyWorld.

1 INTRODUCTION

Dairy consumption varies across cultures. For example, ethnicity affects the liking of yogurt [11]. Moreover, the same dairy products can be perceived as differing in healthiness across cultures [5]. In addition, country characteristics can influence which dairy products are consumed [10]. Therefore, studying the relationship between people and dairy can clarify the broader human relationship with food and deepen the understanding of cultural differences.

Research question. What is the relationship between dairy products and the country of origin?

- For each dairy product, where do they originate from?
- How is the dairy product consumed?
- Is there a relationship between the dairy product and the country of origin?

2 RELATED WORK

Dairy perception and consumption are strongly shaped by culture. At the same time, the history and geography of dairy farming influence how dairy products are produced and valued, and the country of origin of a product can influence consumers. This section goes into more depth on these topics and further discusses food computing and the current research.

2.1 Dairy and culture

A number of studies show that people in different countries perceive and like dairy products in different ways. For example, work on yogurt with different degrees of granularity [11] found that Chinese and Danish consumers responded differently to the presence or absence of particles. Moreover, Johansen et al. [5] studied young Norwegian, Danish, and Californian consumers and showed that all three groups have common motives for consuming calorie-reduced dairy products (e.g., fat content, perceived healthiness and taste), but their relative importance differs across countries. In addition, research on Australian and Chinese consumers [10] reports cross-country differences in the liking of pasteurized milk versus long shelf life milk and in how shelf life labeling affects liking. Lastly, research with a trained panel on flavor lexicons for cheese [2] showed that the panels from New Zealand, Ireland, and the United States used different words to describe the same cheeses, and tended to group cheeses by country of origin.

Beyond sensory perception, cultural and historical factors also influence how dairy is produced, valued, and consumed. For example, in India most milk comes from the buffalo rather than cows because cows have a sacred status in Hinduism. Moreover, in the United States, people are increasingly shifting towards plant-based milks [23]. Lastly, dairy farming can be of great importance to mountain communities [20], family farming cultures [4], and animal welfare [19]. This is further emphasized by Valenze [22], who researched the different religious meanings of milk and the different functions throughout the ages. Thus, dairy is not only a food product, but also has cultural, religious, and socio-economic significance.

2.2 Country-of-origin labeling

Country-of-origin (COO) research has shown that the country where a product is made can influence how consumers evaluate it. This suggests that links between dairy products and countries can shape perceptions and choices. Knight and Calantone [6] propose a model of how COO information is processed cognitively and show that COO effects are complex and cultural context plays an important role in purchase decisions. Moreover, research on the Canadian market [15] found that COO labeling affects the purchase choice and willingness to pay for dairy products. Finally, Yang et al. [25] illustrate how consumer behavior is influenced through COO information after a major food-related scandal.

2.3 Food computing

Food computing and computational gastronomy are emerging research areas that use computational methods to study food-related topics. Surveys and overviews in this area [3, 13] describe how food data (e.g., recipes, images, social media posts) can be analyzed with tools from data science and artificial intelligence to address a wide range of questions about flavor, health, and culinary culture. Several studies focus on culinary cultures and food-country

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relations. For example, Laufer et al. [9] mine cultural relations between different language communities in Europe by analyzing how food cultures are described on Wikipedia pages across different language editions. Moreover, palette varieties around the world are investigated [18] and are found to have strong geographical and cultural similarities between countries. Furthermore, a large visual question answering dataset with text-image pairs is made identifying dish names and their origins [24]. Lastly, two topic modelling approaches are applied to recipes [8, 12] to discover more about the human relationship with food.

2.4 Current research

Previous research has shown that culture influences how dairy products are perceived and consumed, and that the country of origin of a product can affect how consumers evaluate it. Moreover, work on food computing has demonstrated that large-scale online data and computational methods can be used to study food-related topics, including culinary cultures and food-country relations. However, there is little work that combines these perspectives to study how dairy products and countries are linked in textual descriptions. Therefore, this paper uses text mining on Wikipedia articles to investigate how dairy products are associated with their countries of origin.

3 APPROACH

3.1 Data

For this project, Wikipedia pages of dairy products and countries are used. All pages are linked in a Wikipedia page with a list of dairy products[1]. Only the products with all information present (product name, product URL, origin(s), origin URL(s)) are used, resulting in a total of 60 dairy products out of 130. All data is scraped between 3 and 5 January 2026 using the BeautifulSoup library in Python[2] and further processed using pandas [16].

3.2 Method

The linked Wikipedia page provides a list of dairy products and origins, answering the first subquestion. To answer the second subquestion, descriptors are defined to identify processing and preparation methods and serving contexts of dairy products (see Tables 1a and 1b in Appendix B). These descriptors are then used to search on the dairy product pages for paragraphs mentioning one of the keywords, extracting the descriptor, matched keywords and sentence in which the keyword occurred. Each dairy product is then matched with one of the 5 descriptors from each category by assigning the one with the most keyword hits. If multiple descriptors had the highest amount of hits, both descriptors are assigned. The same is done for animal sources of dairy products (see Table 1c in Appendix B), trying to answer the third subquestion. To fully answer the third subquestion, the results from the descriptors are visualized in a world map.

[1]https://en.wikipedia.org/wiki/List_of_dairy_products
[2]https://pypi.org/project/beautifulsoup4/

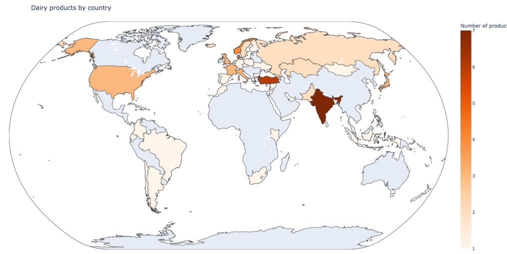


Figure 1: Origins of dairy products

3.3 Validation

For the first subquestion, the results are validated by randomly picking 6 dairy products (10%) and checking their origins with other sources. For the second subquestion, 10% of the extracted descriptor keyword pairs are randomly selected and manually checked with the matched sentences for correctness.

Lastly, for the third subquestion, the country pages are searched for paragraphs containing the dairy product, and the dairy product pages for paragraphs that mention the country of origin. Those paragraphs are extracted and saved in a relations file, used for validation. Since the third subquestion is already partly validated by the validation of the second subquestion, this is only done for 5% of the products.

4 RESULTS

4.1 Product origins

Figure 1 shows the origins of the 60 dairy products from Wikipedia. Darker shades indicate a higher number of products originated from that country. An interactive version is available on GitHub[3]. The interactive version shows on hover which specific products originated from that country. Appendix C describes the mapping procedure.

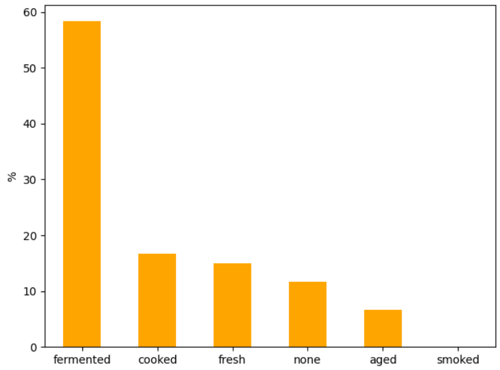
From Figure 1 it can be seen that most countries have just a few products, while Denmark and Norway have 4, Turkey has 6, and India has 7. Most dairy products are from Europe or Asia, while very few come from Africa and Oceania. Note that some products have multiple origins, so totals reflect product–country links rather than unique products.

4.2 Consumption methods

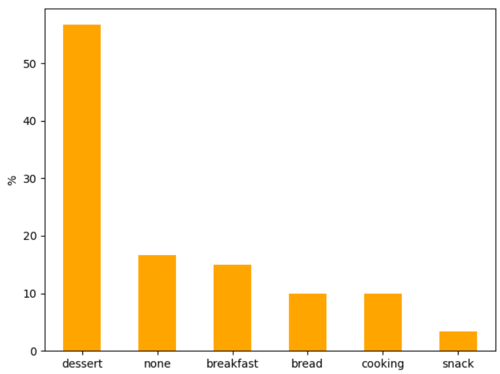
Figure 2 summarizes the descriptors assigned to each of the dairy products after matching the descriptor keywords from Table 1 in Appendix B against the text of the corresponding Wikipedia product pages. For each product, descriptor hit counts were computed, and the descriptor(s) with the maximum number of hits were assigned. Ties therefore result in multiple assigned descriptors and, in turn, the percentages within each plot do not necessarily sum to 100%.

For processing and preparation (Figure 2a), *fermentation*-related keywords were matched most frequently across products. Moreover, for serving context (Figure 2b), most products matched with a *dessert*-related keyword. Lastly, for animal source (Figure 2c) the

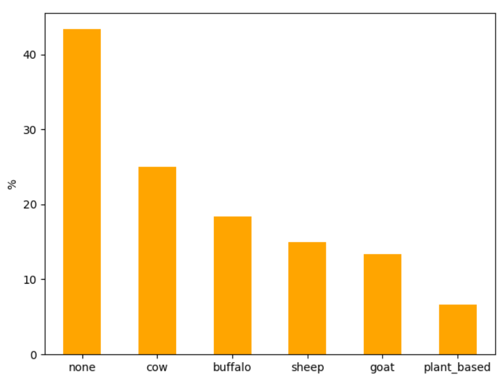
[3]https://dewi-elisa.github.io/dairyWorld/results/world_map.html



(a) Processing and preparation descriptors



(b) Serving descriptors



(c) Animal source descriptors

Figure 2: Descriptor occurrences across dairy products. Note that multiple labels can apply to one product, and therefore the percentages do not sum to 100%.

most frequent outcome was *none*, indicating that many products did not mention a specific animal source in the text.

4.3 Product-country relations

Figure 3 visualizes the geographic distribution of each descriptor by showing, for every country, the percentage of products associated with that country that were assigned the selected descriptor. Note that products can be linked to multiple countries of origin and can be assigned multiple descriptors. Therefore, a single product may contribute to multiple countries and to multiple descriptor maps. Interactive versions of Figure 3 are available on GitHub[4].

The descriptor maps in Figure 3 suggest that descriptors differ in how geographically widespread they are within the dataset. For processing and preparation, *fermented* is the most frequently assigned descriptor (35/60 products; 58.3%) and appears across many countries, whereas *cooked* is less widespread geographically and is most visible in European countries (10/60 products; 16.7%).

For serving context, *dessert* is the dominant descriptor (34/60; 56.7%) and is geographically widespread, while *bread* appears mainly in the United States and Europe (6/60; 10.0%) Moreover, *cooking* appears more strongly in countries around the Mediterranean (6/60; 10.0%).

Lastly, for animal-source descriptors, only 34/60 products (56.7%) contained an explicit animal-source match in the text, meaning that *none* is the most frequent outcome (26/60; 43.3%). Among matched animal descriptors, *cow* is the most common overall (15/60; 25.0%), while *buffalo* (11/60; 18.3%) and *sheep* (9/60; 15.0%) show more geographically concentrated patterns (*buffalo* matches are mainly visible in South America and South Asia, whereas *sheep* matches are most visible in Russia and several Mediterranean countries).

4.4 Validation

To validate if the origins from the Wikipedia page are correct, the internet was searched for different sources mentioning their origins. The results of this can be found in Table 3 in Appendix D. From all validated products, the origin matched with other sources.

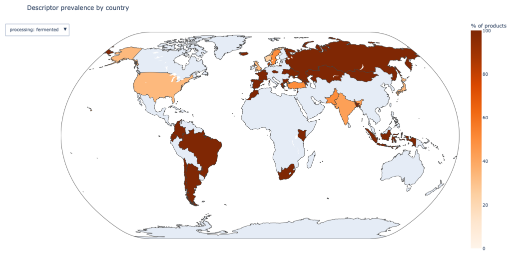
The validation of the results from Figure 2 can be found in Tables 4 to 6 in Appendix D. It can be seen that especially for the processing and preparation descriptors and the serving context descriptors the keywords are often not about the dairy product itself. For example, for yakult the milk is said to be a sweetened milk drink, but this does have nothing to do with yakult being a dessert. For the animal source descriptors, this is already less noisy, but there are still mistakes made. For example, ricotta can be made from *goat milk*, but this also triggered the *oat milk* keyword.

The validation of the results from Figure 3 can be seen in Tables 7 to 9. Multiple country product pairs were chosen with one of their assigned descriptors. These combinations were then validated with the relations file, which provided more context about the product and the country.

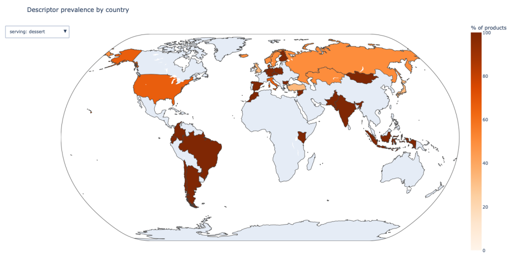
5 DISCUSSION

Manual validation of 10% of the products shows that origin metadata was consistent with external sources. However, descriptor extraction quality varies by category: keyword matches for processing and preparation were frequently not about the product itself (5/12 correct), while serving context (7/11 correct) and animal source (6/8 correct) were more reliable. This suggests that the descriptor distributions and descriptor maps partly reflect Wikipedia phrasing conventions and general food terminology, and conclusions about cultural consumption should therefore be drawn cautiously.

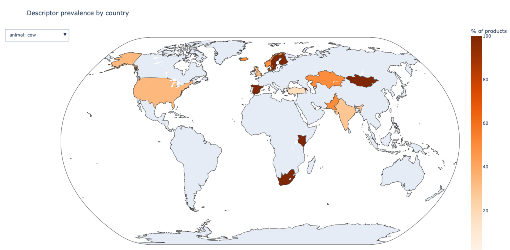
[4]https://dewi-elisa.github.io/dairyWorld/results/descriptor_map_dropdown.html



(a) Processing and preparation descriptors



(b) Serving descriptors



(c) Animal source descriptors

Figure 3: Descriptor prevalence of each descriptor by country.

5.1 Limitations

The current research only used 60 dairy products, while the Wikipedia page consisted of 130 products. Many products were removed because not all metadata fields were complete (e.g., missing origin or missing link to a dedicated Wikipedia page). Some of this information could have been added manually or with more extensive scraping (e.g., because the origin was mentioned in the description column rather than the origin column of the dairy product table). Since the number of products is limited, this can result in unstable country-level patterns, where countries with only one or two products can appear to have very high (or very low) descriptor prevalence in Figure 3.

Furthermore, this research only takes into account textual descriptions from Wikipedia. The data contained ambiguous regional terms, such as Scandinavia, Vikings or Central Asia. Therefore, some compromises were made by mapping ambiguous regional/cultural references to modern countries (see also Appendix C). This may introduce misclassification and affect country-level counts and descriptor prevalence. In addition, food is not only described but also experienced through smell, taste, and texture. The absence of sensory context may bias the descriptors toward what is typically written rather than what is perceived.

Moreover, since the data is scraped from the internet, there might be a bias present to more well-documented countries and regions. Origins that are less well-documented might have additional dairy products which were not found on Wikipedia. This likely contributes to the geographic imbalance visible in Figure 1. Therefore, the map should be interpreted as Wikipedia coverage of products with complete metadata, rather than a complete representation of global dairy diversity or production.

Lastly, it is important to note that the country of origin does not necessarily mean that the product is also consumed there. It might well be that a product originated in country X, travelled to country Y, and became forgotten in country X while country Y is still eating the product. Therefore, not all assumptions about the country of origin and how a product is consumed are warranted.

5.2 Future work

Future work could focus on adding more data sources. When adding additional data sources, data from other modalities could also be used to make up for the absence of sensory context. Additionally, more products could be added, either by additional data sources or by adding missing information to the dairy products that were removed due to missing data.

Moreover, a more in-depth analysis could be made. This could, for example, be done by choosing different descriptors. The current descriptors were handpicked, but another choice could be to make data-driven ones, for example by learning descriptor groups from text embeddings or topic models.

Future work could also incorporate multilingual or non-Wikipedia sources (e.g., regional food databases, cookbooks, or protected designation lists) to reduce English- and Wikipedia-specific bias. Finally, origin could be compared to consumption by incorporating recipe datasets or market data.

6 CONCLUSION

In order to investigate the relationships between dairy products and their country of origin, Wikipedia pages were scraped and used to extract textual relationships with respect to processing and preparation, serving context, and animal source. To extract these relationships, descriptors were defined and matched with the Wikipedia product pages. The origin distribution is concentrated in Europe and Asia, with very few products linked to Africa and Oceania. Descriptor extraction shows that *fermented* and *dessert* are the most frequently matched descriptors, while animal source is often not explicitly mentioned in the text. The country-level descriptor maps suggest that some descriptors are geographically widespread across the dataset (e.g., *fermented*, *dessert*), whereas others show more concentrated patterns (e.g., *cooked* and *buffalo*). Because several countries are represented by only one or two products, country-level percentages can be sensitive to small sample sizes and should be interpreted cautiously. Validation suggests origin metadata is consistent for a sampled subset, but descriptor assignments can be noisy, especially for serving and processing descriptors. Thus, country-level descriptor patterns should be interpreted as Wikipedia-text signals rather than ground-truth consumption behavior.

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A WORK REPORT

In order to find literature, I mainly looked in an online university library. I used queries such as 'milk', 'dairy', 'country of origin dairy', and 'food computing'.

During the programming, I mainly looked at documentation and examples online on websites such as https://www.geeksforgeeks.org and https://www.w3schools.com. One problem I encountered was the Robot Policy of Wikipedia, https://wikitech.wikimedia.org/wiki/Robot_policy#REST_API_rules, which prevented me from scraping the Wikipedia pages. I solved this by adding a user-agent to my requests. For determining the descriptors from Table 1 in Appendix B, I asked ChatGPT-5.2 for keywords. The exact prompt was: *Hi, can you think of descriptors and keywords that say more about dairy? I would like a set of descriptors about: * the processing and preparation of dairy (e.g., fermented, smoked, fresh) * serving context (e.g., breakfast, bread) * the animal source (e.g., goat, cow, sheep, plant) Give about 5 descriptors per category.*

For choosing validation products, I used a random number generator.

B DESCRIPTORS

Table 1: Descriptors and their keywords used to identify processing and preparation methods, serving contexts, and animal sources of dairy products.

(a) Processing and preparation methods

Descriptor	Keywords
Fermented	fermented, fermentation, culture, lactic, probiotic, kefir, yogurt, yoghurt
Aged	aged, ripened, ripening, mature, affinage
Fresh	fresh, unripened
Smoked	smoked, smoky
Heat treated/cooked	cooked, baked, heated, melted, pasteurized, pasteurised

(b) Serving context

Descriptor	Keywords
Breakfast	breakfast, morning, brunch, granola, cereal, oatmeal, porridge
Bread	bread, toast, sandwich, spread, schmear, bagel, cracker, crostini, canape, canapés
Dessert	dessert, sweet, cake, pastry, pastries, pudding, custard, ice cream, gelato, whipped, frosting
Cooking	cooking, sauce, pasta, casserole, gratin
Snack	snack, appetizer, appetiser, cheese board, charcuterie, platter, tapas

(c) Animal source

Descriptor	Keywords
Cow	cow, bovine, jersey, holstein
Goat	goat, caprine, chèvre, chevre
Sheep	sheep, ewe, ovine
Buffalo	buffalo, bufala
Plant-based	plant-based, plant based, vegan, non-dairy, non dairy, dairy-free, dairy free, oat, soy milk, soya milk, almond milk, coconut milk, cashew milk, rice milk, pea milk

C WORLD MAP

In order to be able to make Figure 1, ISO3 mapping was needed for each country of origin. The country converter package [21] was used for this task. However, not all countries were countries: some were cities, regions, former countries, or cultures. Therefore, a manual mapping was made for these cases to a country. The manual mapping to that country is shown in Table 2. For the cities and regions, the country where they are located was chosen. For the former countries, the country that is most closely related to them was chosen. For the Viking culture, the United Kingdom was chosen as the Wikipedia page with dairy product mentions Scotland. When a part of a continent was mentioned, a country central in that part was chosen. Together with product and country names, this was then put in a choropleth map using Plotly [7].

Table 2: Manual mapping of non-country origins to countries for ISO3 conversion

Country	ISO3
USSR	Russia
Khan garh	Pakistan
Scotland	United Kingdom
Vikings	United Kingdom
Central Asia	Kazakhstan
West Sumatra	Indonesia
Scandinavia	Sweden, Norway, Denmark
Alps	Switzerland
Andhra Pradesh	India
Caucasus	Georgia
Punjab	India, Pakistan
Central and Eastern Europe	Poland
Mannheim	Germany
Minoru Shirota	Japan

D VALIDATION

See next page.

Table 3: Dairy products and their reported origins.

Product	Origin	Source (clickable)
Cuajada	Spain	[1]
Fromage frais	France, Belgium	cheese.com
Booza	Syria	Above the Treeline
Mursik	Kenya	[14]
Frozen yogurt	United States	tastingtable.com
Basundi	India	[17]

Table 4: Validation if the processing and preparation descriptor and keyword fits the product.

Product	Descriptor	Keyword(s)	About product?
Haymilk	fermented	fermented	no
Uunijuusto	cooked	baked	yes
Kashk, aaruul, chortan, qurut	fermented	culture, fermentation, fermented, probiotic, yogurt	yes
Kaymak	aged	mature	yes
Dulce de Leche	fermented	yogurt	no
Varenets	cooked	baked, heated, pasteurized	yes
Tzatziki	fresh	fresh	no
Clotted cream	fermented	culture	no
Skyr	fermented	culture, fermentation, yogurt	yes
Kashk, aaruul, chortan, qurut	fresh	fresh	no
Vla	cooked	baked	no
Lassi	fermented	yogurt	no

Table 5: Validation if the serving context descriptor and keyword fits the product.

Product	Descriptor	Keyword(s)	About product?
Dondurma	dessert	ice cream	yes
Kumis	dessert	sweet	no
Gomme	breakfast	porridge	yes
Bulgarian yogurt	cooking	sauce	no
Paneer	dessert	sweet	yes
Yakult	dessert	sweet	no
Mursik	dessert	sweet	no
Ricotta	bread	bread, spread	yes
Fromage frais	bread	bread, spread	yes
Smetana	dessert	cake, dessertm, sweet, whipped	yes
Spaghettieis	dessert	dessert, gelato, ice cream, whipped	yes

Table 6: Validation if the animal source descriptor and keyword fits the product.

Product	Descriptor	Keyword(s)	About product?
Bulgarian yogurt	buffalo	buffalo	yes
Kaymak	buffalo	buffalo	yes
Lassi	buffalo	buffalo	no
Matzoon	buffalo	buffalo	yes
Kaymak	goat	goat	yes
Ricotta	plant_based	oat milk	no
Yayık ayranı	sheep	sheep	yes
Kalvdans	cow	cow	yes

Table 7: Validation if the processing and preparation descriptor and fits the product.

Product	Country	Descriptor	In relations?
Gombe	Norway	cooked	yes
Haymilk	Switzerland	fresh	yes
Leben (labneh)	Lebanon	fermented	yes

Table 8: Validation if the serving context descriptor and fits the product.

Product	Country	Descriptor	In relations?
Soft serve	United States	dessert	yes
Dadiah	Indonesia	dessert	no
Crème fraîche	France	cooking	no

Table 9: Validation if the animal source descriptor and fits the product.

Product	Country	Descriptor	In relations?
Filmjölk	Sweden	cow	no
Kalvdans	Denmark	cow	yes
Paneer	India	buffalo	yes